Shaurya Tanwar

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EDUCATION

Carnegie Mellon University - College of Engineering — **Pittsburgh, PA**

Bachelor of Science in Electrical and Computer Engineering | GPA: 3.50/4.00 — **May 2029**

Relevant Coursework: Introduction to Computer Systems, Principles of Imperative Computation, Intro to Electrical and Computer Engineering, Calculus in 3D, Physics 1 for Engineers

PROFESSIONAL EXPERIENCE

UltraViolet Devices - Engineering Intern | Valencia, CA — **May 2026 - August 2026**

- Implemented TLS 1.2 encryption for device-to-server communications and deployed the update across 100+ production and field devices, securing all data in transit.
- Built a full-stack React/Python analytics dashboard backed by Azure SQL, processing 1,000+ daily logs from 150+ field devices and detecting 10 diagnostic conditions to accelerate troubleshooting for 8 engineers; integrated OpenAI and Google Gemini APIs for automated reporting and natural-language queries.
- Validated GPS and cellular performance across 8 antenna/receiver configurations and 6 operating environments; conducted thermocouple testing of ballasts averaging ~60°C to inform modem, GPS, and power-system component selection for next-generation hardware.

Code Ninjas Sensei - Student Instructor | Valencia, CA — **May 2024 - August 2025**

- Instructed 20+ students (ages 6–17) in JavaScript, Python, Java, Scratch, and Lua through 150 hours of coaching and led coding and robotics camps for 30+ students per session, advancing 15 students by at least 2 proficiency levels.

SuperWIT - Intern | Simi Valley, CA — **May 2024 - August 2024**

- Independently developed and deployed production data visualizations for the Leyland Cypress LLC investment firm website using Next.js, Java, and Python, contributing directly to the firm's live platform: leylandcypress.com

PROJECTS AND ACTIVITIES

CoVM Bytecode Interpreter | C, Valgrind, Stacks, Heaps — **April 2026**

- Implemented a CoVM bytecode interpreter in C using stack- and heap-based runtime structures to execute functions, pointers, arrays, control flow, memory operations, and library calls.
- Built function call frames and low-level memory handling, passing the complete CMU automated test suite and using Valgrind to identify and eliminate memory errors.

Scout Tracker | Java, Android SDK — **June 2023 - August 2024**

- Independently designed and developed "Scout Tracker," a Java-based Android application using the Android SDK, architecting the UI, data model, and core functionality over 200+ hours.
- Built persistent tracking for 125+ rank requirements, 21 merit badges, completion dates, and camping/hiking activity, centralizing long-term advancement records and progress toward Eagle Scout.

Embedded Systems & Circuit Design | I2C, C, Logic Analyzer — **August 2025 - December 2025**

- Built and debugged a multi-device embedded system using I2C communication between a microcontroller, temperature sensor, LCD, and physical inputs, implementing register reads/writes and validating communication with a logic analyzer.
- Built and soldered an AM radio circuit using a hand-wound ferrite-core inductor, capacitors, and variable resistor, tuning circuit components to select and receive radio frequencies.

SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL

Frameworks/Libraries: React, Next.js, Node.js, Android SDK

Tools/Platforms: Git, Linux, AWS, Microsoft Azure

Hardware/Lab: I2C, Oscilloscopes, Logic Analyzers, Multimeters, Soldering, Circuit Prototyping

HONORS

Eagle Scout, Boy Scouts of America — **May 2025**

Gold Presidential Volunteer Service Award — **June 2023**